

Evolution Cheats Imagination: Selection Pressure Amplifies Safety Errors in a Learned World Model

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ABSTRACT

Learned world models let evolutionary search evaluate thousands of controllers cheaply and without physical risk. But selection rewards whatever scores best, including controllers that exploit the model’s errors. We measure this effect for safety. In Walker2d, we evolve residual controllers inside a frozen visual world model and replay every selected controller from the same saved simulator states. In a prospectively locked study (20 paired seeds, four methods, three population sizes, 320 fresh evaluation episodes), increasing search pressure widened the gap between actual and imagined failure by 10.39 percentage points (adjusted 97.5% interval 6.70 to 14.25). Adding calibrated latent noise and a ROSARL-style return-range penalty reduced this widening by 7.22 points (2.49 to 12.18) and cut actual failures at high pressure from 29.56% to 22.36%, retaining 99% of forward progress. Yet the starting controller failed on only 16.88% of the same episodes: the fix limits the damage search does but does not make search beneficial. Exploratory diagnostics on 89 fixed candidates reveal both one-step error and rollout accumulation. A subsequent predictor repair raises failure-endpoint detection from 11.3% to 67.7% on a reused audit split, but its selected controller fails on 3 of 24 branches versus 2 for the starting controller. Better local prediction does not yet yield better policy selection.

CCS CONCEPTS

• **Computing methodologies** → **Bio-inspired approaches**; *Reinforcement learning*.

KEYWORDS

neuroevolution, world models, surrogate models, reality gap, safe reinforcement learning

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1 INTRODUCTION

Evolutionary algorithms need many evaluations. For a robot controller, each real evaluation is slow, and an unsafe candidate may fall over or damage the hardware. A learned world model offers a way out: evaluate candidates inside the model’s predictions instead, where trials are cheap and failure costs nothing [4]. The world model then plays the role of a *surrogate* fitness function [9].

Every surrogate is wrong somewhere, and selection is drawn to exactly those places. If a model wrongly predicts that a risky gait stays upright, that gait scores well, and the best of many candidates is disproportionately likely to be one the model overrates. Decision analysts call this the optimizer’s curse [20]; evolutionary roboticists know it as the reality gap [6, 12]. Ha and Schmidhuber’s controller, evolved in a learned “dream”, scored far higher there than in the real game until they made the dream noisier [4]. For safety, the consequence is specific: the controllers evolution selects look safe in imagination but fail when executed.

This paper asks two questions. *First, how much does selection pressure widen the gap between imagined and actual safety?* *Second, can two simple changes to fitness evaluation reduce that widening without sacrificing task performance?* The changes are Gaussian noise on the model’s latent predictions, scaled to its measured one-step error, and a return-range violation penalty adapted from ROSARL [17]. Neither ingredient is new; our contribution is a controlled measurement of what they do, and fail to do, under evolutionary selection.

We study a Walker2d robot whose controller sees only rendered images. A frozen LeWorldModel (LeWM) [16] predicts how its latent visual state evolves under candidate actions, and separable CMA-ES [19] evolves a residual correction to a frozen starting controller. Every selected controller is replayed in the MuJoCo simulator from the same saved states used in its imagined evaluation (Figure 1), so imagined and actual outcomes are paired.

Our contributions are:

- **A measurement of model exploitation for safety.** In a study locked before any evaluation data were collected, stronger search widened the actual-minus-imagined failure gap by 10.39 percentage points (pp), with adjusted 97.5% interval [6.70, 14.25].
- **A test of a simple mitigation.** Noise plus penalty reduced that widening by 7.22 pp [2.49, 12.18] and lowered actual failures by 7.20 pp at high pressure while retaining 99% of forward progress. Even so, the mitigated controllers failed more often than the starting controller.

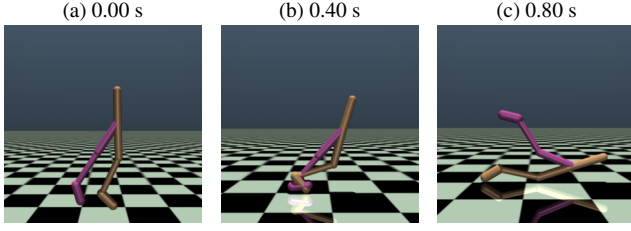


Figure 1: A controller selected by unpenalised, noiseless search, replayed from saved simulator states. The world model predicted no violation over the 0.8 s branch, but the simulator records one at step 62 (0.496 s). This is the first such example for the first search seed at high pressure, chosen for illustration rather than as a representative case.

- **A diagnosis and bounded repair of prediction error.** Exploratory controls separate action feedback, one-step prediction and rollout accumulation. Predictor-only adaptation improves failure-endpoint detection across three fitting seeds, but does not improve the real safety of its selected controller. These follow-ups reuse development episodes and are distinct from the locked main study.

2 RELATED WORK

Reality gaps and surrogates in evolutionary computation. Evolutionary robotics met this problem early. Jakobi et al. showed that controllers evolved in simulation transfer better when the simulator is noisy [6], and the transferability approach adds a learned estimate of how well simulated fitness predicts reality as a further objective [12]. Surrogate-assisted evolutionary computation replaces expensive evaluations with learned models [9], and optimization under noisy or approximate fitness is well studied [10]. Specification gaming, where evolution satisfies the letter of a fitness function rather than its intent, is widely documented [13]. We study the same pressure for a safety constraint, with a learned visual world model as the surrogate.

Evolution and planning in learned models. Ha and Schmidhuber evolved compact controllers inside a recurrent world model and used a stochastic “temperature” to curb exploitation [4]. Recent audits of action-sequence proposals report that larger proposal pools can worsen outcomes under latent prediction error [14]. Guided Safe Shooting plans with MAP-Elites in a learned model under safety constraints [18], and SafeDreamer combines world-model planning with Lagrangian methods [5]. These works plan action sequences or train with gradients; we evolve the parameters of a closed-loop controller and measure how selection pressure changes the safety gap.

Model exploitation in model-based RL. Gradient-based methods guard against exploiting model error by limiting rollout length [7], penalising uncertainty in the reward [21], or sending uncertain states to an absorbing failure state [11]. For constrained problems, CAP inflates predicted costs according to model uncertainty [15]. Our noise is cruder: one fixed scale per latent coordinate, not a state-dependent uncertainty estimate.

Constraint handling and safe RL. Evolutionary computation handles constraints with penalties, feasibility rules and related techniques [2, 3]. Safe RL usually formulates safety as a constrained Markov decision process [1]. ROSARL instead encodes safety in the reward, assigning violations a penalty derived from the range of task values [17]. We use an evolutionary adaptation of that rule; its theoretical guarantees do not transfer to our setting.

3 METHOD

3.1 Task and safety constraint

The robot is the Walker2d model underlying SafetyWalker2d Velocity-v1 [8]. Instead of that benchmark’s velocity cost, we use a *health* constraint: the torso height must stay strictly between 0.8 and 2.0 m and its pitch between -1 and 1 rad. A simulator step lasts 8 ms. We evaluate controllers on short *branches* of 100 steps (0.8 s), each starting from a saved simulator state taken from a recorded episode. A branch *fails* if any of its 100 steps violates the constraint. We disable termination during branches so that post-failure states and progress are still recorded. Throughout, “actual” means execution in MuJoCo, not on physical hardware.

3.2 World model and controller

We reuse a previously trained Walker2d LeWM [16] without modification. Its encoder maps a rendered image to a 192-dimensional latent z . Its predictor maps the last three latents and actions to the next latent, one *block* (10 simulator steps, 80 ms) ahead. Small probes, also frozen, read torso height, pitch and forward speed from a latent.

The *starting controller* is a frozen ensemble of six networks that map visual history to actions, trained by imitation of PPO and PPO-Lagrangian agents. At each block it outputs ten six-motor actions, one per simulator step, so vision runs at 12.5 Hz and actuation at 125 Hz. We evolve only a linear *residual* added to its output: a 60×17 matrix acting on a fixed 16-dimensional projection of the controller’s features plus a bias, giving 1,020 parameters. Actions are clipped to $[-1, 1]$. The residual starts at zero, so the initial search distribution is centred on the starting controller.

3.3 Fitness evaluation in imagination

To evaluate a candidate, we encode a saved start state’s three-frame history and roll the candidate forward for ten blocks in closed loop: the controller acts on predicted latents, and the predictor responds to its actions. Optionally, we perturb each prediction:

$$z_{t+1} = M(z_{t-2:t}, a_{t-2:t}) + k \sigma \odot \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, I). \quad (1)$$

Here σ is the per-coordinate standard deviation of one-step teacher-forced prediction errors (predicted minus encoded latent) on earlier development data. It is a fixed scale, not a state-dependent uncertainty estimate. We compare $k \in \{0, 1\}$. The perturbed latent feeds the controller, the probes and all later predictions.

The probes read the predicted state at every block end. Let $v \in \{0, 1\}$ indicate an imagined health violation and let u be the imagined forward progress accumulated strictly before the first violating block (or over the whole branch if none occurs). We compare two fitness functions, averaged over start states and noise samples:

$$F_0 = \mathbb{E}[u], \quad F_P = \mathbb{E}[u - v(V_{\max} - V_{\min})]. \quad (2)$$

Algorithm 1 Residual evolution in imagination

Require: world model M , starting controller π_0 , population size λ , noise level k , fitness $F \in \{F_0, F_P\}$

- 1: $\mathcal{N} \leftarrow \{\pi_0\}$ ▷ nominees; residual = 0
- 2: initialise separable CMA-ES at residual 0
- 3: **for** $g = 1, \dots, 16$ **do**
- 4: draw 32 fitness start states
- 5: sample residuals $\theta_1, \dots, \theta_\lambda$
- 6: **for** $i = 1, \dots, \lambda$ **do**
- 7: roll out $\pi_0 + \theta_i$ in M with noise k (Eq. 1)
- 8: **end for**
- 9: update $V_{\min}, V_{\max}$; score candidates with F (Eq. 2)
- 10: update CMA-ES; add best candidate to $\mathcal{N}$
- 11: **if** $g \in \{1, 4, 16\}$ **then**
- 12: **select** best member of $\mathcal{N}$ on 64 selection states
- 13: **end if**
- 14: **end for**

Both truncate progress at the first violation; F_P additionally subtracts a penalty equal to the range of task returns observed so far. Following ROSARL [17], the penalty is large enough that no task return can compensate for a violation, and it requires no hand-tuning. $V_{\min}$ and $V_{\max}$ are running extrema of u over the search’s fitness evaluations, updated once per generation before ranking. This is a ROSARL-style adaptation, not the original algorithm.

Noise and penalty act together. Because fitness averages over many start states and samples, it behaves approximately as $(1 - p)\bar{u}_{\text{safe}} - p(V_{\max} - V_{\min})$, where p is the candidate’s imagined violation rate. Noise raises p for behaviour the model cannot predict precisely, and the penalty makes a higher p costly.

3.4 Evolution and selection

Algorithm 1 summarises the search. Separable CMA-ES [19] runs for 16 generations with population size $\lambda \in \{16, 64, 256\}$, using coordinates scaled by the inverse RMS of each projected feature and an initial step size giving an expected unclipped action change of 0.05 RMS. Each generation evaluates all candidates on 32 start states drawn without replacement from 96 *fitness* episodes. Noisy evaluation uses four samples per start state; noiseless evaluation uses one, since it is deterministic. The best candidate of each generation becomes a *nominee* and is rescored on 64 fixed *selection* episodes (16 samples per start state when $k = 1$). At generations 1, 4 and 16 we select the best of the starting controller and all nominees so far. Including the starting controller means that weak search can, and often does, return it unchanged.

4 EXPERIMENTAL DESIGN

Methods and search pressure. Crossing noise ($k = 0$ or 1) with fitness (F_0 or F_P) gives four methods: *control* ($k = 0, F_0$), *penalty*, *noise*, and *noise + penalty*. Each method runs with 20 search seeds, paired across methods so that they share random streams and fitness start states, and with all three population sizes: $20 \times 4 \times 3 = 240$ searches and 720 selected controllers. We define *low pressure* as population 16 at generation 1 (16 candidates evaluated) and

high pressure as population 256 at generation 16 (4,096 candidates evaluated).

Episodes and the locked protocol. Start states come from episodes of competent PPO and PPO-Lagrangian agents with action noise of 0, 0.05 or 0.10, collected until the first violation or 1,000 steps. We take one start state per episode, with enough recorded history and future for a 100-step branch. This conditions start states on the collecting agent surviving around them, so our results do not cover arbitrary initial states. The model, starting controller, noise scale, sample sizes, endpoints and analysis were locked before we collected 480 fresh episodes, split by episode seed into 96 fitness, 64 selection and 320 *evaluation* episodes. All 720 selections were frozen before any evaluation outcome was inspected (Figure 2).

Measurements. Each selected controller, and the starting controller, runs from all 320 evaluation start states twice: in imagination ($k = 0$, and $k = 1$ with 32 samples) and in MuJoCo. *Actual failure* uses the simulator’s health check at every step. *Imagined failure* uses the probes on predicted latents at block ends. For method m at pressure c , the *gap* is

$$D_{m,c} = \bar{v}_{m,c}^{\text{actual}} - \bar{v}_{m,c}^{\text{imag}}, \quad A_m = D_{m,H} - D_{m,L}, \quad (3)$$

with imagined failure measured at the method’s own noise level. A_m , the *amplification*, is how much stronger search widens the gap.

Hypotheses and statistics. We declared two co-primary contrasts: $A_{\text{control}} > 0$ (search widens the gap) and $A_{\text{noise+penalty}} - A_{\text{control}} < 0$ (the combined method reduces that widening). A smaller gap could simply mean more imagined failures, so we separately required, at high pressure, that the combined method reduce actual failure relative to control (95% interval below zero) while keeping at least 90% of its mean forward progress (lower 95% bound of the ratio above 0.90). Intervals come from 10,000 paired bootstrap replicates that resample search seeds and evaluation episodes independently, preserving pairing across methods and pressures. The two primary intervals have 97.5% coverage each (Bonferroni family coverage 95%); all other intervals are descriptive 95% intervals. The 6,400 seed–episode pairs per cell are not independent: the independent units are 20 seeds and 320 episodes.

5 RESULTS

5.1 Selection pressure widens the safety gap

Both primary hypotheses were supported (Figure 3). For the control, the gap grew from 18.70% at low pressure to 29.09% at high pressure, an amplification of +10.39 pp [6.70, 14.25]. Imagined failure barely moved (0.31% to 0.47%), while actual failure rose from 19.02% to 29.56%: stronger search found controllers that looked equally safe to the model but failed more often. Figure 4 shows the full grid. Larger budgets generally worsened control safety, but the primary test compares only the declared endpoints and does not assert monotonicity.

5.2 Noise and penalty reduce actual failures

Noise + penalty reduced the amplification by 7.22 pp [2.49, 12.18] relative to control. This was not merely a matter of predicting more failures: at high pressure, the combined method failed on 22.36% of

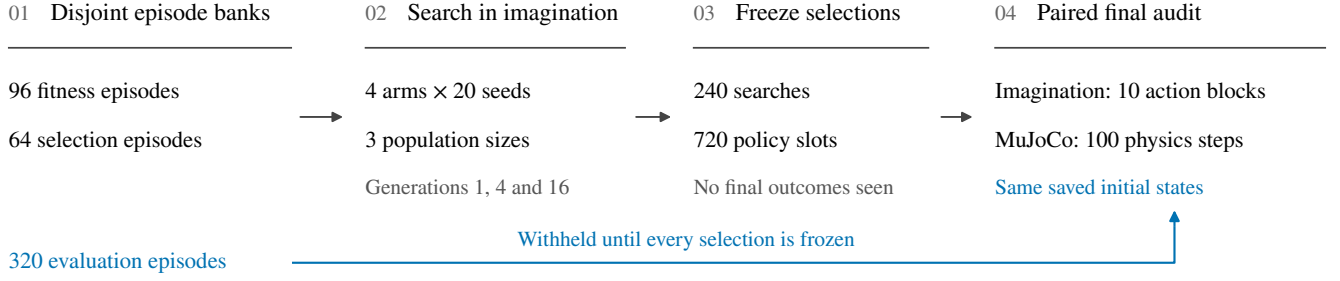


Figure 2: Locked study design. Episodes are disjoint across roles, with one start state per episode. All selections were frozen before final evaluation. In imagination, controllers act on predicted latents; in MuJoCo, on newly rendered images. Paired start states therefore do not imply identical action sequences.

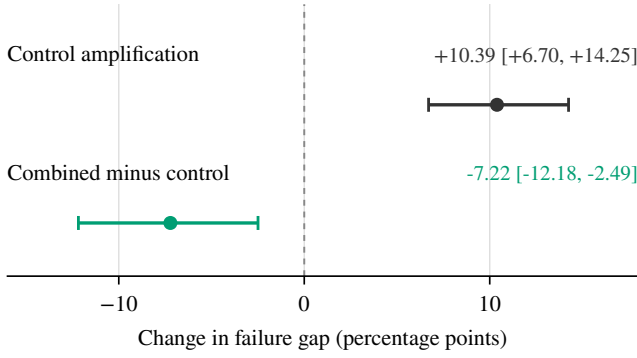


Figure 3: The two co-primary contrasts with 97.5% intervals. Positive amplification means stronger search widened the actual-minus-imagined gap; the negative change means noise + penalty reduced that widening.

evaluations versus 29.56% for control (Table 1), a paired difference of -7.20 pp $[-10.98, -3.70]$. Its mean forward progress was 2.146 m versus 2.167 m, a ratio of 0.990 $[0.982, 0.998]$. Both practical criteria were met, at the cost of a small (about 1%) loss of progress.

Descriptively, both ingredients contributed. At high pressure, adding the penalty changed actual failure by -2.09 pp $[-5.39, 0.59]$ without noise and by -3.30 pp $[-6.16, -0.48]$ with it; adding noise changed it by -3.91 pp $[-7.81, -0.05]$ without the penalty and by -5.11 pp $[-8.55, -2.02]$ with it. These secondary intervals are unadjusted and do not establish an interaction. Noisy search also queries the model four times as often per candidate, so we have not isolated the effect of noise at equal model compute.

5.3 The starting controller remains safer

The comparison that matters most for deployment is with the controller search started from. That controller failed on 54 of the 320 evaluation episodes (16.88%). The best evolved method still failed 5.48 pp $[2.56, 8.44]$ more often. Even at low pressure, the control was already 2.14 pp $[0.67, 4.05]$ worse than the starting controller. Noise and penalty therefore reduce the harm that optimization

Table 1: Outcomes at high pressure (population 256, generation 16). Imagined failure uses each method’s own noise level; the starting controller’s uses $k = 0$. Evolved rates average 20 seeds on 320 shared episodes. Progress includes post-failure motion.

Method	Actual (%)	Imagined (%)	Progress (m)
Control	29.56	0.47	2.167
Penalty	27.47	0.47	2.162
Noise	25.66	5.09	2.155
Noise + penalty	22.36	4.95	2.146
Frozen baseline	16.88	0.31	2.141

causes; they do not make optimization helpful. The imagined failure rates in Table 1 are also far below the actual ones (4.95% versus 22.36% for the combined method), so even the noisy model is badly overconfident.

5.4 Where the gap comes from

We split each gap into three parts using two additional readings of every actual branch: the health check at block ends only, and the probes applied to encoded *real* images at block ends. (i) Violations that occur and recover between block ends contribute only 0.38 pp to the control’s high-pressure gap. (ii) The probes on real images over-report violations by 7.27 pp, which narrows rather than widens the gap. (iii) The remainder, the difference between the probes on real and on imagined latents, is +35.98 pp (Figure 5). The readout detects more violations on real images than on imagined latents. This segment-level decomposition motivates testing the predicted dynamics, but neither establishes perfect readout accuracy at failure onset nor separates local prediction error from rollout accumulation. We test those distinctions below.

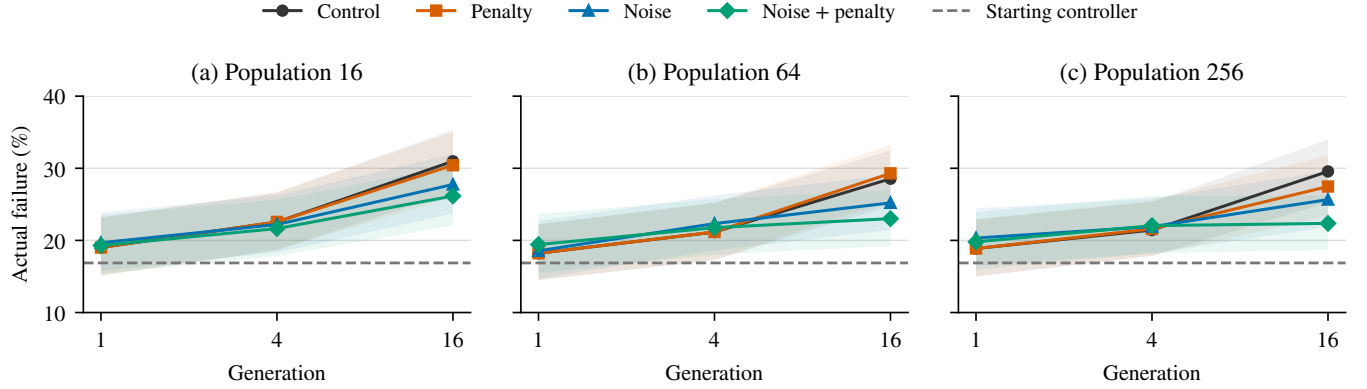


Figure 4: Actual failure across all population sizes and generations, with descriptive pointwise 95% intervals. The dashed line is the starting controller (16.88%). All curves share the same 320 evaluation episodes.

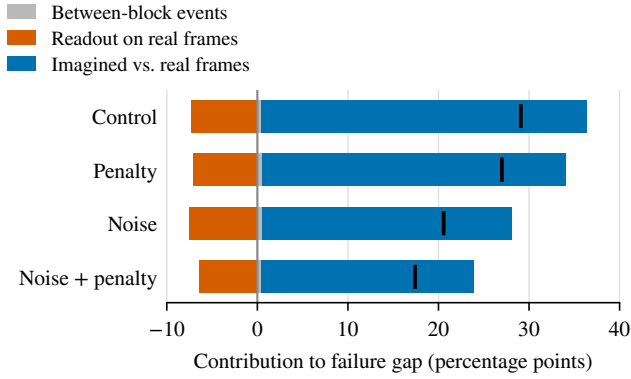


Figure 5: Decomposition of the high-pressure actual-minus-imagined gap. Black ticks mark the net gap. Nearly all of it arises between probe readings on real and imagined latents; the probe over-reports violations on real images.

6 EXPLORATORY DIAGNOSIS AND MODEL REPAIR

6.1 Candidate quality

A natural next step is to check shortlisted candidates in the simulator before choosing one. In a separate development study (two paired seeds per method), this simulator-verified selection always returned the starting controller, because no shortlisted candidate failed strictly less often. That raised a more basic question: does search produce good controllers that imagination fails to recognise, or are good controllers simply rare near the starting point?

Design. We froze a pool of 89 distinct controllers before any new simulator queries: the starting controller; the four imagined winners of those development searches; 32 outcome-blind samples from their populations; 32 random residuals (eight directions at four scales); and 20 contracted or capped versions of the winners. Capping clips the difference between the evolved and starting actions

on the same input at every block. All 89 were scored in imagination, on 32 simulator-selection episodes, and then, after all choices were frozen, on 64 separate check episodes. These episodes had been used in earlier development, so this audit is exploratory: its intervals are unadjusted and its candidates are correlated.

Findings. On the check episodes, the starting controller failed on 4 of 64. Of the 88 other candidates, 86 failed more often, one tied, and one failed less (3 of 64). That one candidate had only tied on the selection episodes, so no predeclared rule would have chosen it. The candidate chosen by simulator selection failed on 6 of 64, rescuing none of the starting controller’s failures and adding two. The candidate chosen by both imagined rankings failed on 12 of 64. Imagination could not separate the candidates: without noise, *all* 89 had zero imagined failures, and with noise the rank correlation between imagined and actual failure was 0.04.

Figure 6 shows a clear pattern instead. The further a candidate moved the starting controller’s actions, the more often it failed (descriptive Spearman correlation 0.78 across this correlated pool). Toned-down and capped winners failed less than uncapped ones but still more than the starting controller. For the four controllers whose traces we decomposed (the starting controller, the one improved candidate, and the two simulator-selected choices), every actual failure was visible at a block end, and the probes on real images detected every one of them somewhere in the segment, with two to five false alarms each. Imagination without noise predicted none. This is *eventual* detection on four controllers; it does not establish accurate detection at the first failing endpoint across the complete pool.

6.2 Local prediction error and rollout accumulation

We replayed the saved trajectories through the original predictor, holding the executed real actions fixed and refreshing its three-frame context from real latents every one, two, five, or ten blocks. Original closed-loop imagination provides a fifth path. The refreshes and future action tapes are privileged retrospective controls.

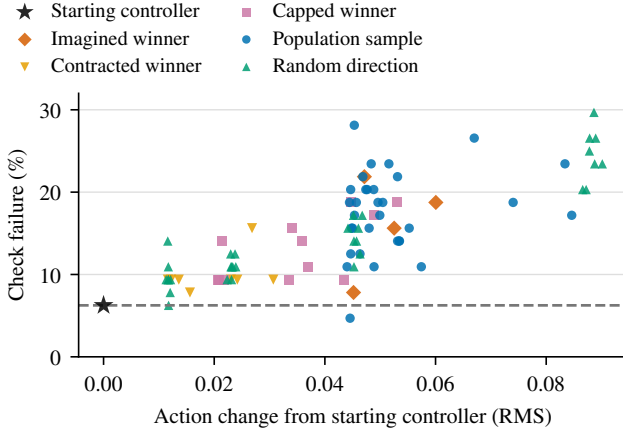


Figure 6: Exploratory audit of 89 fixed controllers on 64 reused check episodes. Failure rises with how far a candidate changes the starting controller’s actions on the same inputs. The dashed line is the starting controller.

They require no new simulator interaction and are not deployable root-time predictions.

On the 64 reused check episodes, 861 candidate/episode pairs failed densely. The block containing the first failure ended unsafe in 819 pairs; 42 transient events were no longer unsafe at that endpoint. Closed-loop imagination detected 0 of those 819 endpoints, and replaying the exact real actions without refresh detected only 1. Refreshing real history every block detected 128 (15.6%), versus 12.3% and 2.2% with refresh every two and five blocks. Different imagined actions therefore do not explain the misses, and frequent refresh improves predictions without resolving their local error.

The original probe on real images detected 572/819 first unsafe endpoints (69.8%), versus 799/861 failed segments eventually (92.8%). Thus the readout also has an onset-detection limitation. These endpoint measurements can follow the first dense event by up to 72 ms; they are not advance-warning rates. Residuals against real-image probe readings exceeded 0.05 m in height or 0.1 rad in pitch before nearly all failures, but also somewhere in 71.5% of non-failing trajectories with one-step real-history refresh. Residual divergence alone is not a validated failure detector.

6.3 Predictor repair improves detection, but not controller selection

Frozen design. We assigned the same 96 previously exposed source episodes to four disjoint groups of 24 by a fixed hash of episode identity: probe fitting, predictor fitting, development, and audit. All 89 candidate branches from an episode stay together. The audit is excluded from fitting and selection in this procedure, but is reused development evidence rather than fresh confirmation. We compared the original pipeline, probe-only repair, predictor-only repair, and both repairs, with three fitting seeds restarted from the original weights. The encoder, latent projector, action encoder, normalisers and all running statistics remained frozen. Only the

predictor and its output projector, or the existing same-capacity probe network, received updates.

Each seed used 1,000 AdamW updates per component, batch 256, with predictor/probe learning rates $10^{-5}/10^{-4}$, weight decay 10^{-4} and gradient-norm clipping at one. Half each batch sampled the last two available endpoints through a failure-containing block; the other half sampled remaining eligible endpoints. No later post-failure state was fitted. Probe loss was physical MSE scaled by (0.2 m, 0.5 rad, 5 m/s). Predictor loss was latent MSE plus 0.1 times that physical loss through the original frozen probe. Targets use detached real latents and privileged simulator labels.

The predeclared one-step development gate required at least a 10 pp gain in first-unsafe-endpoint detection, no more than a 2 pp increase in false alarms at still-healthy endpoints, and at most 1.25× baseline healthy-endpoint MAE for each of height, pitch and speed. Passing arms were ranked by detection, then false alarms; the choice was frozen before audit summaries. All intervals below are descriptive 95% bootstraps over whole source episodes, conditional on this candidate pool; fitting seeds and candidate branches are not independent episodes. Point gates are engineering criteria, not significance tests.

Prediction results. Only predictor-only repair passed development, and it also passed audit (Figure 7). On the audit group’s 344 unsafe first-failure endpoints, mean detection rose from 11.3% to 67.7%, a paired gain of 56.4 pp [48.0, 61.2]. The three fitting seeds achieved 69.2%, 66.0%, and 68.0%. False alarms at endpoints strictly preceding dense failure, including all endpoints on non-failing branches, rose from 0.39% to 0.85%. Healthy height/pitch MAE decreased from 0.02375 m/0.07903 rad to 0.02250 m/0.07571 rad; speed MAE increased from 0.4462 to 0.4824 m/s. Mean height bias at the failure-containing endpoint fell from +0.1488 to +0.0273 m. Without real-history refresh, real-action-tape detection improved from 0% to 31.2%, leaving substantial rollout error. Probe-only repair achieved 0.9% one-step detection; combined repair reached 37.3% but failed the speed-error retention gate. Better regression of real-image states did not make that probe reliable on predicted latents.

Controller selection. We then applied the same imagined-only nomination rule to the original model and the development-selected repaired predictor: require strictly lower predicted failure than the starting controller and at least 90% of its predicted progress, breaking ties by progress then candidate index. Otherwise retain the starting controller. All nominations preceded numerical audit summaries. The original model predicted no baseline failures and fell back to the starting controller. The repaired model nominated candidate 076, whose predicted failure was 12.5% versus the starting controller’s 18.1%. On the separate audit group, this candidate failed on 3/24 branches versus 2/24 for the starting controller, retaining 98.3% of actual progress. The paired failure difference was +4.17 pp [0, 12.5]. This small reused audit does not establish general worsening, but provides no evidence of a controller improvement. No controller parameters were retrained and no nominee was promoted.

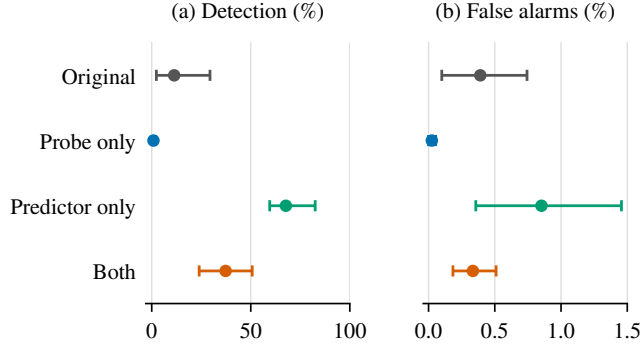


Figure 7: Exploratory repair audit: 24 reused source episodes, 89 fixed candidates. Points average the three fitting seeds for repaired arms; whiskers are source-episode bootstrap 95% intervals. Detection is at the first unsafe block endpoint, not an advance warning. False alarms use still-healthy endpoints. Predictor-only repair passes the frozen gate.

7 DISCUSSION AND LIMITATIONS

What the results show. Evolution in a learned world model behaves like optimization against any imperfect surrogate: more selection finds more of the surrogate’s errors, and for safety those errors are false reports of safe behaviour. Noise and a return-range penalty push back measurably, but leave a substantial gap. The retrospective controls expose both one-step errors and their accumulation. Targeted predictor adaptation improves local failure prediction, yet the resulting model still misidentifies a beneficial controller in the tested pool. Prediction accuracy and decision quality require separate evaluation.

Implications. First, a smaller imagined-versus-actual gap is not evidence of a safer controller: the gap can shrink because imagined failures rise. Evaluation must include actual outcomes. Second, results should be compared with the starting controller, not only with unmitigated search; against that reference, every method in the main study made things worse. Third, repairing local prediction can improve detection without improving selection. The next test should measure controller-risk rankings over complete closed-loop rollouts, their stability across source episodes and fitting seeds, and the headroom available in the fixed pool before another evolution run.

Limitations. The main study used one environment, one frozen world model and one starting controller, on 0.8 s branches from start states conditioned on survival of the collecting agent. The starting controller itself fails on a sixth of branches, and earlier readiness and transfer screens in this project failed; we make no claim of a safe controller. Development choices, including $k = 1$, were informed by earlier pilots, although the final episodes and analysis were fixed in advance. We did not test independently pretrained model seeds, full episodes, other environments, state-dependent uncertainty, or evolution against the simulator itself. The three repair fitting seeds share the same pretrained model. All follow-ups in Section 6 reuse development episodes and a correlated candidate pool; their point gates and unadjusted intervals are not fresh confirmation or safety

Table 2: Main-study compute. A predictor row is one latent transition for one sample. Final evaluation covers selected controllers, the starting controller and reference action sequences.

Phase	Simulator steps	Predictor rows
Episode collection	463,921	0
Fitness evaluation	0	344,064,000
Selection	0	22,195,200
Final evaluation	23,104,000	78,144,000
Total	23,567,921	444,403,200

guarantees. Some upstream costs (agent training, earlier failed runs) are not fully accounted for.

Cost. Table 2 separates collection, search and evaluation costs. The main study ran in 6.8 hours on one GPU with no gradient updates. The main-study ledger records 26,838,972 simulator steps across 24 preserved runs, plus 3,600,058 shared collection steps for the world model and probes; it is not a lifetime total. An independent audit verified all 118 frozen source files and 11,322 query archives and reproduced the analysis from saved outputs. The later candidate-pool audit added 854,400 simulator steps and 768,960 predictor rows. The divergence diagnostic used 427,200 predictor rows. Repair used 768,000 predictor fitting rows, 768,000 probe fitting rows, 3,000 optimiser updates per component across three seeds, and 320,400 inference predictor rows in 97.2 s including setup and writes. The divergence and repair follow-ups used zero new simulator steps, renders or image encodes; engineering tests and earlier interaction costs are separate. The repair audit verified 534 inference archives and three checkpoints. The approved private archive is not an anonymous public replication package.

8 CONCLUSION

Evolution exploits the errors of a learned world model, and for safety this means selecting controllers that look safe but are not. In a locked Walker2d study, stronger search widened the actual-minus-imagined failure gap by 10.4 percentage points. Calibrated latent noise and a ROSARL-style penalty reduced that widening and cut actual failures by 7.2 points at almost no cost in progress, but every evolved method remained less safe than the controller it started from. Exploratory follow-ups reveal local prediction error, rollout accumulation, and imperfect readout at failure onset. Predictor-only repair raises local endpoint detection substantially but does not improve the real safety of the selected controller. The next milestone is reliable ranking and selection of controller risk, followed by fresh confirmation and evaluation across longer episodes, other robots and independently pretrained models.

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